USN [ ] [22EC42]

# FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

## FIELDS AND WAVES

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | | | | |
|---|---|---|---|---|---|
| | | **PART-A** | **05 x 02=10** | | |
| **1.** | **a.** | Infinite uniform line charges of 5 nC/m lie along the (positive and negative) x and y axis in free space. Find E at P(0,0,4) | L3 | CO1 | **02** |
| | **b.** | Define Potential Gradient. | L1 | CO2 | **02** |
| | **c.** | State Ampere's Circuital law. | L1 | CO4 | **02** |
| | **d.** | Summarize the differential form of Maxwell's Equation. | L2 | CO5 | **02** |
| | **e.** | Compute propagation constant in free space for a wave with 100MHz. | L3 | CO6 | **02** |
| | | **PART-B** | **06 X 05=30** | | |
| **2.** | **a.** | Determine an approximate value for the total charge enclosed in an incremental volume of $10^{-9}$ $m^3$ located at the origin, if $D = e^{-x} \sin y\, a_x - e^{-x} \cos y\, a_y + 2z\, a_z$ C/m$^2$. | L3 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Use vector form of Coulomb's law by locating a charge $Q_1=3 \times 10^{-4}$C at M(1,2,3) and a charge of $Q_2 = -10^{-4}$ C at N(2,0,5) in a vacuum. Determine the force exerted on $Q_2$ by $Q_1$ | L3 | CO1 | **05** |
| | | | | | |
| **3.** | **a.** | Given the potential field, $V = 2x^2y-5z$, and a point P (−4, 3, 6), to determine several numerical values at point P: the electric field intensity and the volume charge density $\rho_v$. | L3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Derive an expression for potential field of a point charge. | L3 | CO2 | **05** |
| | | | | | |
| **4.** | **a.** | The field $B = -2a_x + 3a_y + 4a_z$ mT is present in free space. Determine the vector force exerted on a straight wire carrying 12 A in the a AB direction, given A (1, 1, 1 ) and B ( 2, 1, 1 ). | L4 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Find H at P(4,6,10) in cartesian coordinates for an infinitely long current filament passing through the origin and point A(0,0,10). The current of 25A is directed from origin to A. | L3 | CO3 | **05** |
| | | | | | |
| **5.** | **a.** | Explain any two applications of ampere's circuital law | L3 | CO4 | **05** |

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-562112

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| | **b.** | Explain Faraday's law of electromagnetic induction. | L3 | CO4 | **05** |
| | | | | | |
| **6.** | **a.** | Derive Maxwell's second equation. | L3 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Select the value of k so that the following pairs of fields satisfy Maxwell's equations in the region where conductivity and volume charge density is zero. $E=(kx-175t)a_y$ V/m, $H=(x+35t)a_z$ A/m, Permeability=0.35 H/m , Permitivity = 0.01 F/m. | L3 | CO5 | **05** |
| | | | | | |
| **7.** | **a.** | Derive the electromagnetic wave equation. | L3 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Explain in detail about the wave propagation in free space. | L3 | CO6 | **05** |
| | | | | | |
| | | **PART-C** | **06 X 10=60** | | |
| **8.** | **a.** | Derive an expression for electric field intensity at a point P lying in XY plane due to an infinitely long conductor lying in z axis | L3 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Three infinite uniform sheets of charge are located in free space as follows:<br>3 nC/m$^2$ at z = −4, 6 nC/m$^2$ at z = 1, and −8 nC/m$^2$ at z = 4.<br>Determine E at the point:<br>**i.** $P_A(2, 5, -5)$;<br>**ii.** $P_B(4, 2, -3)$;<br>**iii.** $P_C(-1, -5, 2)$;<br>**iv.** $P_D(-2, 4, 5)$. | L3 | CO1 | **10** |
| | | | | | |
| **9.** | **a.** | Given the nonuniform field $E = y\,a_x + x\,a_y + 2\,a_z$ .Calculate the work expended in carrying 2 C from B (1, 0, 1) to A (0.8, 0.6, 1) along the shorter arc of the circle $x^2+ y^2= 1$; z = 1. | L3 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | In free space, let $D = 8xyz^4a_x + 4x^2z^4a_y + 16x^2yz^3\,a_z$ pC/m$^2$. Determine<br>**i.** the total electric flux passing through the rectangular surface z = 2, 0 < x <2, 1 < y < 3, in the $a_z$ direction.<br>**ii.** E at P (2, −1, 3). | L3 | CO2 | **10** |
| | | | | | |
| **10.** | **a.** | From the Biot-Savart's law, derive the expression for magnetic field intensity at a point P and distance R from the infinitely long straight current carrying conductor. | L3 | CO3 | **10** |

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| | **b.** | The point charge Q = 18 nC has a velocity of $5 \times 10^{6}$ m/s in the direction $a_v = 0.60a_x + 0.75a_y + 0.30a_z$. Calculate the magnitude of the force exerted on the charge by the field:<br>**i.** $B = -3a_x + 4a_y + 6a_z$ mT ;<br>**ii.** $E = -3a_x + 4a_y + 6a_z$ k V/m | L3 | CO3 | **10** |
| | | | | | |
| **11.** | **a.** | Derive the Maxwell's Equation from Gauss law and Faraday's law. | L3 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Explain in detail about modified Ampere's Circuital law. | L3 | CO4 | **10** |
| | | | | | |
| **12.** | **a.** | Demonstrate the wave equations for a perfect dielectric. | L3 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | Document the Maxwell's equations in both point form and integral form for both steady state fields and time varying fields. | L3 | CO5 | **10** |
| | | | | | |
| **13.** | **a.** | A 10 GHz uniform Plane propagates through free space has an amplitude 15V/m. Calculate velocity of propagation, Phase Constant, Wavelength, charatericstic impedance of the medium and amplitude of H. | L3 | CO6 | **10** |
| | | **OR** | | | |
| | **b.** | The time domain expression for the magnetic field of a plane wave travelling in a non-magnetic medium is given as $H(x,t)=0.2\cos(6\pi \times 10^{8} t - 10.2x)a_z$ $A/m$. Find direction of wave propagation, dielectric constant, intrinsic impedance, Expression for electric field and time average power density. | L3 | CO6 | **10** |